Sahaj Bhadja

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Summary

Software and systems engineer with 2+ years in **requirements-driven testing**, **vehicle-network (CAN) verification**, autonomous-driving **simulation**, and test automation, turning requirements into specifications, tests, and **reports**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Defined **validation** and reconciliation checks for production data pipelines across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs at 99%+ reliability with documented, CI/CD-gated releases.
- Restructured a fault-tolerant async job system with explicit state transitions and **failure isolation**, processing 1K+ jobs and tripling throughput under concurrent load while preventing cascading task failures.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*

- Built **closed-loop simulation** evaluation for an autonomous-driving behavioral planner trained on 100K+ trajectory logs, analyzing off-road and lane-following failures and improving long-horizon trajectory planning by 18% in simulation.
- Designed a spatial-temporal encoder fusing HD-map road geometry with ego-vehicle trajectory history through multi-head attention, improving closed-loop stability 15% and **comparing results** against Wayformer-style and MotionLM-style baseline encoders.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Automated nightly Jenkins **validation** that built executables, installed them on test VMs, and ran integration tests across Windows, Linux, and macOS configurations, producing CSV reports and failure logs.
- Translated **requirements** into tested data-sharing microservices from analysis through deployment, coordinating Agile sprints with Microsoft and GitHub stakeholders and improving response times 30% and pipeline efficiency 68%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Ran unit, integration, and end-to-end Selenium **HMI tests** in Jenkins before Docker staging releases, achieving 95%+ build validation while Sentry, Azure Monitor, and Kibana surfaced production faults.
- Built a MERN resource-management platform with WebSocket real-time updates across eight microservices, synchronizing 2M+ records across locations and using analytics to cut resource wastage 42% and raise lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Captured client requirements and **specified** an end-to-end system architecture with 12+ integration points, presenting clear design reviews to client stakeholders and reducing design errors 25% before implementation began.
- Improved **cross-platform** compatibility 15% for Android, iOS, and web clients of a house-help service booking app, holding sub-200ms API latency across 50K+ records with Firebase real-time status synchronization.

Skills

Vehicle Systems: CAN Bus, DBC Files, SocketCAN, Frame Integrity (E2E), Closed-Loop Simulation, Autonomous Driving
Requirements & Testing: Requirements Analysis, Requirements Traceability, Specification Writing, Test Planning, HMI Testing
Test Methods: Integration Testing, Regression Testing, End-to-End Testing, Cross-Platform Testing, Fault Injection, Selenium
Languages & Analysis: Python, C++, Java, SQL, Statistical Analysis, Data Analysis, PyTorch, Weights & Biases, Hydra
Tools: Git, Jenkins, Linux, Docker, CI/CD, JIRA, PyTest, Agile/Scrum, Root-Cause Analysis
Communication: Technical Documentation, Test Reporting, Design Reviews, Stakeholder Communication, Cross-Functional Teams

Projects

BusVerify - Requirements-Driven CAN Verification | *Python, CAN, DBC, SocketCAN, opendbc*

- Built a requirements-driven **CAN** harness on real Honda DBC layouts, running 1,425 simulated drives across trim variants and injected defects and forecasting a 1.1% FMVSS 111 camera-latency miss rate on premium trims.

CPU-scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented and **tested** FCFS, SJF, SRT, Round Robin, priority, and MLQ scheduling across preemptive and non-preemptive configurations with Gantt visualization, reducing average turnaround time 25% and debugging time 20%.

CaptionCraft - Image Caption Generator Live | *Python, PyTorch, Transformers, Flask*

- **Benchmarked** fine-tuned BLIP, GIT, ViT-GPT-2, and a custom vision-language model on a 10K-image, 100K-caption dataset, analyzing results to raise caption accuracy by 15 points, from 81% to 96%.